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## **Laboratory Work 1: Organizational Data Governance and Policy Automation**

**Course: Management of Organizational Data**

**Theme: Organizational Data Governance Frameworks, Compliance, and Data Stewardship**

**Programming Language:** Python 3.10+

Tools: `pandas`, `numpy`, `re` (regular expressions)

### **Objective**

The goal of this lab is to operationalize enterprise data governance concepts. You will design a governance RACI matrix, perform data quality validation, enforce privacy compliance rules (GDPR and Kazakhstan Personal Data Protection Law), and auto-generate an audit report using Python.

**Deadline – 26 September 2026, defend and upload python files, report**

Part 1: Theoretical and Strategic Framework (30% of total mark)

#### Task 1: RACI Matrix Design

An e-commerce company in Kazakhstan is introducing a new customer data collection pipeline. Construct a RACI Matrix for the following 5 tasks across the 5 organizational roles listed below:

Roles:

1. Chief Data Officer (CDO)
2. Lead Data Steward
3. Data Engineer
4. Legal and Compliance Officer
5. Marketing Manager

### Tasks to Assign:

1. Defining enterprise Data Quality rules for customer email and phone formats.
2. Approving data access permissions for third-party marketing vendors.
3. Implementing automated PII (Personally Identifiable Information) anonymization in data pipelines.
4. Auditing cross-border data transfer compliance under GDPR and KZ PDP laws.
5. Resolving duplicate customer master records in the database.

Deliverable: Present a table mapping each Task to the 5 Roles using R (Responsible), A (Accountable), C (Consulted), or I (Informed). Ensure exactly one A per task.

## Part 2: Practical Implementation in Python (70% of total mark)

### Task 2: Data Quality and Regulatory Compliance Engine

Prepare dataset, with a raw CSV dataset containing dirty customer records (`raw\_customer\_data.csv`). You must build an automated Python script that enforces Data Governance policies.

#### Step 1: Dataset Setup

Create a script or notebook that loads the following sample data:

```
```python
import pandas as pd
import numpy as np

data = {
    "customer_id": [101, 102, 103, 104, 105, 106],
    "full_name": ["Arman Aliyev", "Elena Petrova", "John Doe", "Aisha Saparova", "Dmitry Ivanov", "Sanzhar N"],
    "email": ["arman@kbtu.kz", "elena.p@gmail.com", "invalid-email-at-com", "aisha@domain.org", np.nan, "sanzhar@company.kz"],

```

```

    "phone": ["+77015551234", "87071112233", "+77029998877", "12345",
"+77053334455", "77080001122"],

    "age": [24, 17, 35, 140, 29, -5],

    "kz_iin": ["000115501234", "070420609876", "12345", "950812300111",
"880315400222", "990101300000"],

    "consent_given": [True, True, True, False, True, False]
}

```

```
df = pd.DataFrame(data)
```

## Step 2: Implement Governance Validation Functions

Write Python code to execute the following governance controls:

1. Completeness Control: Identify and count records with missing critical fields (`email`, `full\_name`, `kz\_iin`).
2. Data Quality and Validity Checks:
  - Email Validation: Verify emails using a regex pattern (`r"^[w\.-]+@[w\.-]+\.\w+\$"`). Mark non-matching emails as invalid.
  - Age Sanity Check: Age must be a valid human age (\$18 ≤ age ≤ 120). Flag underage (<18) or out-of-bound values.
  - Kazakhstan IIN Validation: Valid IINs must be strictly 12 numeric digits (`^d{12}\$`).
3. Privacy and Compliance Policy Enforcement (GDPR and KZ Law):
  - Consent Filtering: If `consent\_given` is `False`, the customer's record must not be processed for marketing. Filter these out.
  - Data Anonymization (Pseudonymization): Write a function `anonymize\_name()` to mask full names for compliance (e.g., "Arman Aliyev" → "A. Aliyev").

## Step 3: Generate Automated Audit Report

Calculate and print an Executive Data Governance Report containing:

Total records audited.

Overall Data Quality Score (Percentage of records passing all validity checks).

Count of consent-revoked records isolated.

A summary dictionary of flagged quality issues.

Expected Output Format

Your script should display an output structured similar to this:

```
=====
DATA GOVERNANCE and COMPLIANCE AUDIT
=====
Total Input Records Audited : 6
Consent Revoked (Isolated) : 2
Valid Records Approved    : 2

--- Quality Control Failures ---
Missing Email Count      : 1
Invalid Email Format      : 1
Invalid Age Values       : 2
Invalid KZ IIN Strings   : 1
Overall Data Quality Score : 33.33%
=====
```

### Part 3: Critical Thinking and Discussion Questions

Answer the following short-answer questions in your final report:

1. Data Localization: Article 12 of Kazakhstan's Personal Data Protection Law requires that personal data of KZ citizens be physically stored within the country. How does this law impact an enterprise deciding between storing user databases on-premises vs. using global public clouds (e.g., AWS, Microsoft Azure)?
2. Data Stewardship: If a database contains duplicate entries for the same customer (e.g., `"Arman Aliyev"` and `"A. Aliyev"`), explain the difference between what the Data Owner does and what the Data Steward does to resolve it.

### Submission Requirements

File format: A single Jupyter Notebook (`.ipynb`) or zipped Python script (`.py`) along with a PDF report for all parts